Keigo Kusumegi

PhD Candidate, Information Science, Cornell University

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Research Interests

My research interests center on understanding *how artificial intelligence is reshaping society*, with a particular focus on *scientific innovation*. To address this, I apply a range of computational approaches, including machine learning, econometrics, and network science, to large-scale datasets.

Education

Cornell University Aug 2023 – Present (expected May 2028)
PhD in Information Science. Advisor: Yian Yin *Ithaca, NY*

University of Tsukuba Apr 2021 – Mar 2023
M.S. in Policy and Planning Sciences *Ibaraki, Japan*

University of Tsukuba Apr 2017 – Mar 2021
B.S. in Policy and Planning Sciences *Ibaraki, Japan*

Publications

Journal publications

1. **Kusumegi, K.**, Yang, X., Ginsparg, P., De Vaan, M., Stuart, T., & Yin, Y. Scientific production in the era of large language models. *Science*, 390, 1240–1243 (2025).
2. **Kusumegi, K.**, Acuña, D. E., & Sano, Y. Dissecting the gender divide: Authorship and acknowledgment in scientific publications. *Scientometrics* (2026).
3. **Kusumegi, K.** & Sano, Y. Dataset of identified scholars mentioned in acknowledgement statements. *Scientific Data*, 9, 461 (2022).

Conference papers

4. **Kusumegi, K.**, Yang, X., Ginsparg, P., De Vaan, M., Stuart, T., & Yin, Y. Scientific production in the era of large language models. *Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), AI4Science Track* (2026).

Preprints

5. **Kusumegi, K.**, Yang, X., Ginsparg, P., De Vaan, M., Stuart, T., & Yin, Y. A robust association between LLM use and scientific productivity: Assessing stopping-time selection. *arXiv* (2026).
6. **Kusumegi, K.** & Sano, Y. Citations and gender diversity in reciprocal acknowledgement networks. *arXiv* (2021).

Working papers

7. **Kusumegi, K.**, Wang, Y., & Yin, Y. The specialization of interdisciplinary innovators in science and technology.

Research & Professional Experience

Cornell University Aug 2023 – Present
Graduate Research Assistant *Ithaca, NY*

Cornell University Aug 2024
Text Analysis Consultant, Summer Graduate Fellowship in Digital Humanities *Ithaca, NY*

University of Tsukuba Dec 2019 – Mar 2023
Undergraduate & Graduate Research Assistant, Sano Lab *Ibaraki, Japan*

G-RIPS Jun 2021 – Aug 2021
Research Intern *Sendai, Japan*

- Designed and implemented an optimization algorithm for wireless base station placement using topology and graph theory.

OPTiM Corporation Sep 2020 – Oct 2020
Research Intern *Tokyo, Japan*

- Developed a CNN-based multi-object detection model for industrial quality control.

MARK Creative Jan 2019 – Jul 2023
Software Engineer *Ibaraki, Japan*

- Designed and implemented a graph database (Neo4j) and its API for data analysis and system optimization.
- Built an automated programming-question generator for Python exercises.

Selected Media Coverage

- *Nature*: How AI slop is causing a crisis in computer science
- *Brookings*: The train has left the station: Agentic AI and the future of social science research
- *The Atlantic*: Science Is Drowning in AI Slop
- *Ars Technica*: New OpenAI tool renews fears that "AI slop" will overwhelm scientific research
- *The Conversation*: What the hyperproduction of AI slop is doing to science
- *Cornell Chronicle*: AI gives scientists a boost, but at the cost of too many mediocre papers
- *Phys.org*: Scientists who use AI tools are publishing more papers than ever before

Awards & Recognition

- Funai Overseas Scholarship ($120K) 2023
- QUAD Fellowship ($50K); one of 100 selected from over 3,200 applicants 2023
- University of Tsukuba President's Commendation 2023
- Tsukuba Graduate Study Scholarship 2023
- Nakagawa Ikueikai Scholarship 2021
- Best Poster Award, 16th Network Ecology Symposium 2021
- Scholarship, Japan Society for the Promotion of Science 2018

Reviewing

Nature (2025 ×1) Nature Communications (2025 ×1) KDD (2026 ×3)

Mentoring

- Nichole Zheng, Undergraduate, Cornell University 2025–2026
- Angela Xu, Undergraduate, Cornell University 2025

Invited Talks

1. Japanese Science of Science Workshop *Tokyo, Japan / Jun 2026*
2. Complex Systems Research Exchange *Online / Dec 2025*
3. Japanese Science of Science Workshop *Tokyo, Japan / Oct 2023*
4. 17th Network Ecology Symposium *Online / Mar 2022*

Conference Presentations

1. KDD'26: AI4Science Track — *Jeju, South Korea / Aug 2026*
2. IC2S2: International Conference on Computational Social Science — *Burlington, VT / Jul 2026*
3. Complex Networks 2025 — *Binghamton, NY / Dec 2025*
4. KDD'25 Workshop: LLMs for Scientific and Societal Advances — *Toronto, Canada / Aug 2025*
5. ICSSI: International Conference on the Science of Science — *Copenhagen, Denmark / Jun 2025*
6. NetSci2023 — *Vienna, Austria / Oct 2023*
7. Conference on Complex Systems 2022 — *Palma de Mallorca, Spain / Oct 2022*
8. Reconstructing Network Dynamics from Data — *Los Angeles, CA / Aug 2022*
9. Socioeconomic Networks and Network Science Workshop — *Online / Aug 2022*
10. NetSci2022 — *Online / Jul 2022*
11. Complex Networks 2021 — *Online / Nov 2021*
12. Science of Innovation and Success Workshop — *Tokyo, Japan / Aug 2021*
13. IC2S2: 7th Intl. Conference on Computational Social Science — *Online / Jul 2021*
14. Networks 2021: Joint Sunbelt and NetSci Conference — *Online / Jul 2021*
15. 16th Network Ecology Symposium (Best Poster Award) — *Online / Mar 2021*
16. 7th Service Engineering Symposium — *Online / Feb 2021*
17. Network Seminar 2020 — *Online / Dec 2020*
18. NetSci2020: 7th Satellite on Quantifying Success — *Online / Sep 2020*